

Enhancing Air Quality Prediction with LLM-Based Explanation and Decision Support

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Abstract—Accurate air-quality forecasting provides utility only when its outputs can be interpreted and translated into actionable decisions. This work introduces a decision-support pipeline that first forecasts hourly PM2.5 concentrations from meteorological and pollution history, then maps the numerical predictions to the Air Quality Index (AQI), and finally employs a large language model (Qwen2.5-7B-Instruct) to generate grounded explanations and user-oriented health recommendations. The forecasting component implements the mRMR-RF-based feature selection and ISSA-LSTM-based prediction design proposed in recent air-quality forecasting literature. Experimental evaluation uses the Beijing PM2.5 hourly dataset from UCI with a 60-hour lookback window and chronological train-validation-test splits. The best numerical benchmark achieved was AR(24) with an RMSE of $21.67 \mu\text{g}/\text{m}^3$, followed by persistence ($22.02 \mu\text{g}/\text{m}^3$), a baseline LSTM from literature ($24.53 \mu\text{g}/\text{m}^3$), and the ISSA-LSTM ($28.78 \mu\text{g}/\text{m}^3$). Although ISSA improved certain training behaviors, it did not outperform simple time-series baselines on this test split, underscoring the critical role of benchmark comparison in environmental forecasting. The LLM component generated 100 forecast explanations and 25 interactive question-answer responses. A rule-based hallucination audit revealed 0.0% hard hallucinations across five grounding checks: PM2.5 number match, AQI match, category consistency, supplied-feature grounding, and physical-range validity. Thus, the primary contribution is not merely a prediction model but a reproducible framework for converting environmental forecasts into accountable, auditable language-based decision support.

Index Terms—Air Quality Prediction, Air Quality Index (AQI), PM2.5 forecasting, Long Short-Term Memory (LSTM), Large Language Models (LLMs)

I. INTRODUCTION

Air pollution forecasting is a time-sensitive prediction problem with direct implications for health behavior, transportation planning, school activities, and municipal advisories. PM2.5 is especially important because fine particles can penetrate deeply into the respiratory system and are routinely used in public air-quality communication [1]. In practice, however, a qualitative forecast (for example, $\text{PM}_{2.5} = 106.9 \mu\text{g}/\text{m}^3$) is not self-explanatory for most users. The forecast must be translated into a category, risk interpretation, and practical recommendation.

Deep Learning (DL) architectures are attractive for air pollution forecasting because PM2.5 has strong temporal structure. Hourly air pollution reflects emissions, atmospheric stability, boundary-layer dynamics, wind, humidity, and lagged pollutant concentration. Long Short-Term Memory networks (LSTMs) were designed to learn long-range dependencies through gated recurrent cells [2], and they have been widely adopted for environmental time series. Recent work by Wu et

al. [1] proposed an air-quality prediction architecture that combines mRMR-RF feature selection with an ISSA-optimized LSTM. That architecture motivated our base model.

The present work extends numerical forecasting with an LLM explanation and decision-support layer. The key design constraint is that the LLM must not invent pollutant values, unsupported causes, or medical claims. The system therefore passes a compact structured context to the LLM and audits the returned explanations against deterministic checks. This turns the LLM from an unconstrained text generator into a post-processing component whose outputs can be verified against the same forecast record that produced them.

This paper primarily focuses on analysis of the LSTM framework for air pollution forecasting and LLM-based decision support while reporting both successes and negative findings. The strongest numerical result came from simple autoregression (AR) and persistence, not from the tuned LSTM. This is scientifically important because, on a strongly autocorrelated PM2.5 sequence, a complex model must beat naive temporal baselines.

The explanation layer, by contrast, demonstrated high grounding under the implemented checks, but still showed qualitative weaknesses in some answers, primarily overly generic health advice. This motivates the need for fine-tuning language models instead of relying solely on prompting.

II. RELATED WORK

Hybrid models that combine feature selection with recurrent forecasting have become a staple in air-quality prediction. Wu et al. [1], for instance, proposed a framework using mRMR-RF feature selection alongside ISSA-LSTM. Their core argument? Reducing input redundancy and tuning LSTM hyperparameters can boost AQI forecasts. Their design fuses a filter criterion, a tree-based importance estimate, and a swarm-intelligence optimizer. Our approach follows a similar blueprint, but we test it on Beijing’s PM2.5 hourly data and explicitly pit it against simple baselines. Speaking of LSTM: first introduced by Hochreiter and Schmidhuber [2], it tackled the vanishing-gradient problem in recurrent networks. Thanks to its memory cell and gates, it can retain information over long sequences—a handy trait when pollutant levels hinge on past weather and emission patterns. Random Forests, credited to Breiman [3], deliver robust nonlinear modeling and variable importance via ensembles of randomized trees. In our work, RF serves more for interpretability and feature ranking than as the main predictor. Moving to mRMR: Peng, Long, and Ding [4] formalized this mutual-information method, aiming

for features that are highly relevant to the target yet low in redundancy. Then there’s the Sparrow Search Algorithm (Xue and Shen [5]), a swarm optimizer mimicking sparrow foraging and anti-predator behavior; its ISSA variants tweak search dynamics to better balance exploration and exploitation. LLMs have recently gained traction as interfaces for structured models, particularly when users crave plain-language explanations. Qwen2.5, for example, is a family of open LLMs with instruction-tuned versions at scales like 7B [6]. That said, LLMs can generate plausible but unsubstantiated claims if prompts and outputs aren’t reined in. Hence, our system relies on schema-constrained JSON generation and post-hoc hallucination tests, rather than taking fluency as proof of correctness. Finally, AQI category mapping follows the U.S. EPA technical assistance document for daily air quality reporting [7].

III. RESEARCH OBJECTIVES AND CONTRIBUTIONS

The study addresses the following research questions:

- **RQ1:** How do existing ISSA-LSTM and baseline LSTM models compare with persistence and 24-lag AR model?
- **RQ2:** Which input variables dominate next-hour PM2.5 forecasting under mRMR and RF importance analysis?
- **RQ3:** Can an LLM produce useful AQI explanations while remaining grounded in the forecast record?
- **RQ4:** What qualitative failure modes remain after hard hallucination checks pass?

The primary contributions of this paper are as follows:

- A reproducible PM2.5 forecasting pipeline using 60-hour sequences, chronological splits, MinMax scaling, LSTM training, and ISSA hyperparameter search.
- A feature-selection and interpretation layer combining mRMR selected features with Random Forest impurity importance.
- A schema-constrained LLM layer that converts predicted PM2.5 and AQI category into explanation, recommendation, and confidence note including a deterministic hallucination audit with five hard checks and one soft confidence signal.
- An analysis of LLM answers for practical user questions such as running, children playing outside, cycling with a mask, window ventilation, and school sports.

IV. DATASET AND PREPROCESSING

The experiment uses the Beijing PM2.5 dataset from the UCI Machine Learning Repository. The dataset contains hourly PM2.5 measurements from the U.S. Embassy in Beijing and meteorological measurements from Beijing Capital International Airport [8]. The dataset spans 2010-01-01 through 2014-12-31.

After cleaning, the working dataset contains 43,824 hourly rows. The original PM2.5 column contained 2,067 missing rows, representing 4.72% of the dataset. Missing PM2.5 values were imputed by forward fill followed by backward fill. The categorical wind-direction field was not used in the final model.

The retained numerical features were PM2.5, dew point, temperature, pressure, and wind. All features were scaled to [0,1] with MinMax scaling for LSTM training. The target is next-hour PM2.5 concentration in physical units after inverse transformation.

Chronological splitting was used to avoid temporal leakage, resulting in 31,510 sequences for training, 3,501 for validation, and 8,753 for testing. Each supervised sample used a 60-hour lookback window.

TABLE I
DESCRIPTIVE STATISTICS FOR RETAINED INPUT FEATURES

Feature	Mean	Std. Dev.	Min	Max
pm25	97.801	91.376	0.000	994.000
dew	1.817	14.433	-40.000	28.000
temp	12.449	12.199	-19.000	42.000
pressure	1016.448	10.269	991.000	1046.000
wind	23.889	50.011	0.450	585.600

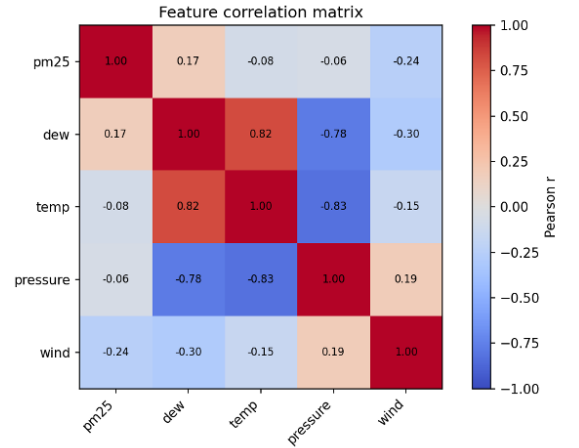


Fig. 1. Feature correlation matrix. Dew point and temperature are strongly positively correlated, while both are negatively correlated with pressure.

V. METHODOLOGY

A. Feature Selection

The feature selection component has two roles: reducing redundancy for the predictor and summarizing drivers for the LLM. mRMR was used to greedily select variables that are informative about next-hour PM2.5 while avoiding highly redundant variables. In this work, the top three selected features were PM2.5, pressure, and wind.

A supporting Random Forest (RF) was trained for feature-importance analysis. Its out-of-bag R^2 was 0.926021. The impurity importance was dominated by PM2.5 at 94.46%, while dew point, wind, temperature, and pressure each contributed roughly 1–2%.

TABLE II
RANDOM FOREST FEATURE IMPORTANCE

Feature	RF Importance	Importance (%)
pm25	0.9446	94.4604
dew	0.0159	1.5933
wind	0.0138	1.3785
temp	0.0129	1.2949
pressure	0.0127	1.2729

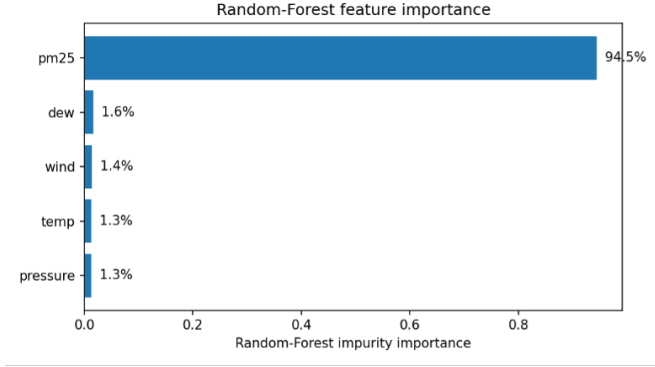


Fig. 2. Random-Forest feature-importance profile, showing the dominance of lagged PM2.5.

B. Forecasting Models

The baseline LSTM uses two stacked LSTM layers with 64 hidden units, dropout 0.2, MSE loss, Adam optimization at learning rate 10^{-3} , and batch size 64. Inputs are 60-hour windows of all five scaled features, and the output is scaled next-hour PM2.5.

The ISSA-LSTM uses the same supervised formulation but tunes hidden size, number of layers, dropout, learning rate, and batch size. The search used 8 sparrows and 6 iterations, with a one-epoch validation-MSE probe as the fitness function.

Two naive baselines were included:

- Persistence predicts next-hour PM2.5 as the current PM2.5.
- AR(24) is an ordinary least-squares autoregression on the PM2.5 series with 24 lags.

C. LLM Explanation and QA Layer

The explanation model is Qwen2.5-7B-Instruct. The LLM receives only the predicted PM2.5 value, calculated AQI integer, AQI category, and the supplied top feature values. It is instructed to return output with six answers:

- predicted_pm25_ugm3
- predicted_aqi
- aqi_category
- primary_driver
- explanation
- health_recommendation
- confidence_note

The hallucination detector applies five hard checks and one soft check:

- H1 verifies PM2.5 number match.

- H2 verifies AQI match.
- H3 verifies AQI category consistency.
- H4 rejects invented pollutants or unsupported features.
- H5 checks physical-range validity.
- H6 flags low token log probability as a soft confidence concern.

VI. EXPERIMENTAL RESULTS

A. Forecasting Metrics

Table III reports held-out chronological test performance.

TABLE III
TEST-SET FORECASTING METRICS

Model	RMSE	MAE	Bias	R^2	Pearson r
Persistence	22.02	11.82	0.01	0.945	0.972
AR(24)	21.67	11.90	0.03	0.947	0.973
LSTM baseline	24.53	14.85	-8.05	0.932	0.973
ISSA-LSTM	28.78	19.09	-14.91	0.906	0.970

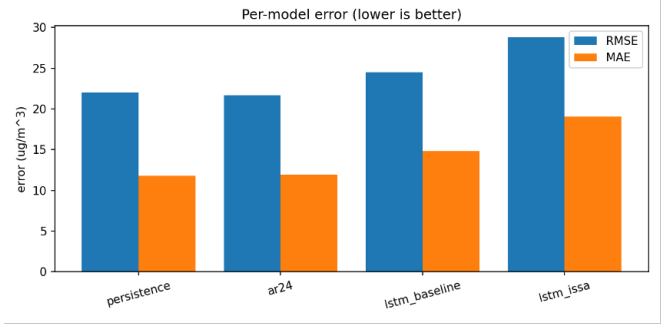


Fig. 3. Per-model error comparison. Lower RMSE and MAE are better.

The most important observation is that AR(24) and persistence outperform both LSTM variants on RMSE and MAE. The ISSA-LSTM underpredicts more strongly and obtains the weakest RMSE and category accuracy.

B. Training Dynamics and Prediction Shape

The baseline LSTM loss curve shows rapid improvement during the first four epochs and gradual flattening afterward. The validation loss reaches its best value near the end of training, suggesting limited overfitting within 10 epochs.

Prediction overlays show that both LSTM variants capture the timing of many pollution episodes but systematically smooth extreme peaks. This smoothing behavior is common for MSE-trained regressors because large spikes are rare relative to moderate observations.

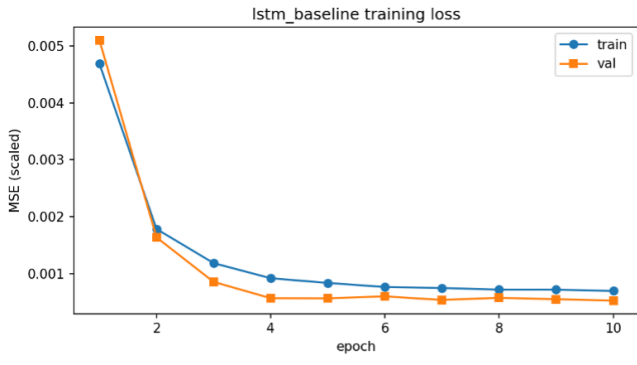


Fig. 4. Baseline LSTM training and validation loss.



Fig. 5. ISSA-LSTM training and validation loss.

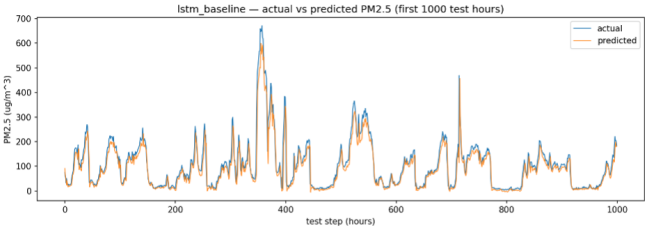


Fig. 6. Baseline LSTM actual-versus-predicted PM2.5 for the first 1000 test hours.

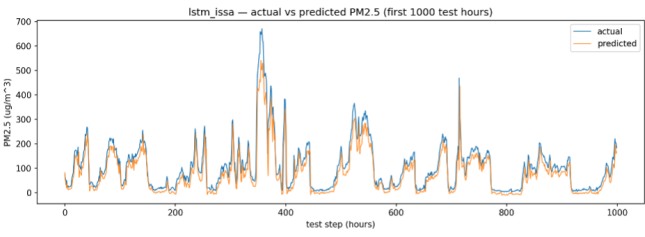


Fig. 7. ISSA-LSTM actual-versus-predicted PM2.5 for the first 1000 test hours.

C. AQI Category Analysis

The AQI category distribution reveals that the predictions overproduce Unhealthy and Moderate cases while underrepresenting Good cases. Actual Good cases total 886, but predicted

Good cases are only about 381. This is consistent with the model bias toward midrange values and its difficulty at the low end. The model also underpredicts some hazardous and very unhealthy cases, which is more serious because these cases require stronger warnings.

Per-category errors show a different pattern from aggregate metrics. The best model has relatively low RMSE in the Good and Moderate ranges but poor R^2 within those categories because the within-category variance is small. Hazardous cases have high RMSE but a stronger Pearson correlation, indicating that the model tracks high-pollution variation but underestimates magnitude. For decision support, this means category calibration matters as much as point prediction.

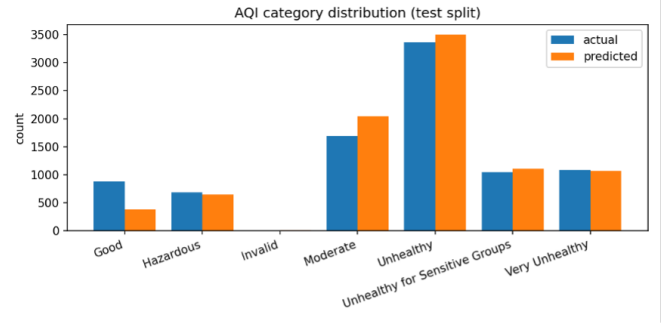


Fig. 8. Actual and predicted AQI category distribution on the test split.

TABLE IV
PER-AQI-CATEGORY ERROR INTERPRETATION

Category	n	RMSE	MAE	Bias	MAPE	Interpretation
Good	886	8.15	5.67	+5.35	77.64	Low absolute error
Moderate	1690	13.45	7.31	+4.58	34.65	Upward shift cases
USG	1045	19.02	9.53	+3.91	21.12	Boundary confusion
Unhealthy	3367	18.66	11.90	-0.10	13.02	Balanced bias
Very Unhealthy	1083	30.25	18.12	-5.81	9.63	Peak smoothing
Hazardous	682	42.02	25.17	-14.21	7.29	Largest errors

VII. LLM EXPLANATION RESULTS

The LLM produced 100 explanation attempts for test predictions. The mean average log probability was -0.062, and the hallucination audit reported 0 hard hallucinations.

TABLE V
HALLUCINATION-AUDIT SUMMARY

Metric	Value
Number of explanations	100
Hard hallucinations	0
Hallucination rate (%)	0.0
PM2.5 number match H1 (%)	100.0
AQI integer match H2 (%)	100.0
Category bracket H3 (%)	100.0
Feature grounding H4 (%)	100.0
Physical range H5 (%)	100.0
Low-confidence H6 (%)	0.0
Mean average log probability	-0.062

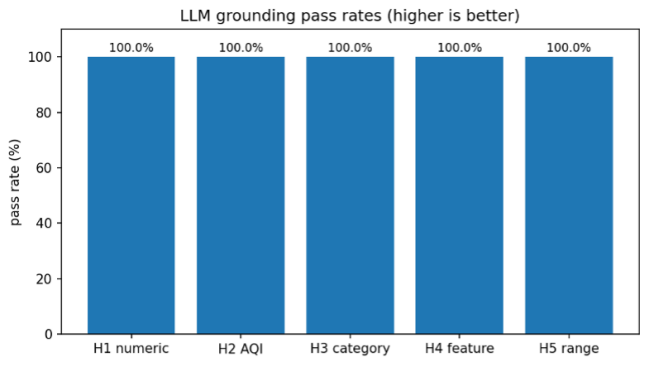


Fig. 9. LLM grounding pass rates across the five hard checks.

The checks ensure that generated text is consistent with the forecast record, though they do not prove that every recommendation is optimal medical advice.

VIII. TWO-PAGE ANALYSIS TABLE OF INTERACTIVE LLM ANSWERS

Table IX expands the QA outputs into a qualitative analysis table. It is intentionally long enough to occupy approximately two pages in the manuscript because the QA component is central to the decision-support contribution. The table shows that the same forecast record can support different user-facing recommendations depending on the question. It also shows where the LLM should be improved: answers should be more explicit about uncertainty, ventilation tradeoffs, mask recommendations, and institutional thresholds.

IX. DISCUSSION

One central quantitative finding stands out: simple temporal baselines remain difficult to surpass in next-hour PM2.5 forecasting. Persistence and AR(24) benefit from PM2.5’s strong autocorrelation. While LSTM models captured general trends, they smoothed peaks and underpredicted high-pollution episodes. From a research standpoint, this is valuable—it prevents overstating the benefits of optimization. Future models should incorporate direct multi-step forecasting, probabilistic intervals, loss functions that penalize underprediction of high AQI events, and calibration against category thresholds.

Feature-selection results clarify why persistence is so dominant. Random forest importance allocates about 94.5% of impurity importance to lagged PM2.5. Meteorological features do matter, but their incremental contribution is small in next-hour forecasts. That doesn’t mean weather is irrelevant—its effect is already partially baked into the current PM2.5 state and shows up over longer timescales. For longer horizons, wind, pressure, dew point, and temperature could become more influential.

The LLM layer works best as an interface, not a standalone source of truth. It turns model outputs into activity-specific guidance, and its JSON structure enables automatic validation. The zero hard-hallucination rate suggests numeric fields and supplied-feature constraints worked well for this run. But a

closer look reveals limitations: some raw outputs have formatting defects, some recommendations feel generic, and some explanations overplay a meteorological feature even when RF importance is driven by PM2.5. A production system should pair the LLM with stricter templates for high-risk categories.

For public health, the forecast model and LLM are best viewed as advisory. AQI thresholds are population-level communication tools—individual risk depends on age, respiratory or cardiovascular disease, exertion level, indoor filtration, and exposure duration. So the system should include confidence notes and steer clear of implying medical diagnoses. The most defensible output? A grounded statement: “The model predicts Unhealthy AQI; reduce prolonged outdoor exertion, especially for sensitive groups.”

X. THREATS TO VALIDITY

- Dataset scope: the experiment uses one city and one historical period. Performance may change in other cities, sensor networks, climates, and policy regimes.
- Imputation assumptions: forward/backward fill is simple and reproducible, but it can smooth missing pollution episodes and may affect lag-based models.
- Evaluation horizon: the target is next-hour PM2.5. Results should not be generalized to day-ahead or week-ahead forecasting without new experiments.
- LLM audit coverage: H1-H5 detect numeric and feature-grounding violations, but they do not fully evaluate medical appropriateness, readability, or policy compliance.
- Prompt dependence: the LLM results depend on the exact JSON schema, decoding settings, and post-processing. Different prompts may yield different failure modes.
- Optimization budget: ISSA used a small population and limited iterations. A larger search might improve performance, but it would also raise computational cost.

XI. FUTURE WORK

When modeling moving forward, researchers should prioritize calibrating their models to accurately capture extreme pollution episodes. Several promising techniques stand out: quantile regression, asymmetric loss functions, temporal convolutional networks, transformer-based time-series architectures, and explicit weighting for Very Unhealthy and Hazardous categories. To boost robustness, ensemble forecasts that blend AR models, persistence, LSTM networks, and tree-based approaches offer a viable path.

On the explanation front, future efforts need to assess whether users truly understand and trust the system—ideally through human studies. The current approach of auditing for hallucinations is a necessary first step, but it doesn’t go far enough. A more robust evaluation would score each recommendation on factuality, completeness, readability, specificity, and actionability. Additionally, the QA system should embed local policy rules—for example, thresholds for canceling school sports or indoor-air guidance based on available filtration.

Implementing a retrieval-augmented framework could anchor health recommendations in a vetted AQI guidance source. This would minimize generic or incomplete advice and let the system attach citations to its outputs. However, careful management is essential: retrieval must be version-controlled, validated, and kept distinct from forecast values to prevent conflating model uncertainty with guideline uncertainty.

XII. CONCLUSION

We built and tested a research prototype for air-quality prediction that incorporates LLM-based explanations and decision support. Numerical forecasting results highlighted the importance of baseline comparisons. AR(24) and persistence models outperformed both LSTM variants on our test split. Notably, the ISSA-LSTM underpredicted severe pollution episodes. Interpretability analysis revealed that PM2.5 history dominates next-hour predictions, while meteorological variables add only modest context. Under a deterministic grounding audit, the LLM layer translated forecasts into structured explanations and user-oriented recommendations. Across 100 explanations, it achieved a 0% hard hallucination rate. The key takeaway: LLMs can add value to environmental forecasting when they are constrained, audited, and used solely as a communication layer—not as the forecasting engine itself.

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APPENDIX A REPRODUCIBILITY SUMMARY

TABLE VI
REPRODUCIBILITY DETAILS FOR THE REPORTED EXPERIMENT.

Item	Value
Run ID	45030900
Device	cuda
Random seed	42
Lookback window	60 hours
Train/validation/test	31,510 / 3,501 / 8,753
LLM model	Qwen2.5-7B-Instruct
Explanations generated	100
Interactive QA pairs	25
Hard hallucination rate	0.0%

The supplementary files used to generate this paper include prediction tables, LLM explanation logs, hallucination flags, QA examples, and rendered figures. The report intentionally preserves negative results because they indicate where the pipeline requires additional modeling before deployment. In particular, the ISSA search should be expanded and coupled with early stopping, event-weighted validation, and a comparison against tuned ARIMA, gradient-boosted trees, and transformer time-series baselines.

APPENDIX B PROMPT CONTRACT AND HALLUCINATION-AUDIT DETAILS

Designing the explanation module hinged on a foundational contract: the LLM could paraphrase, add context, and suggest actions, but it could never alter the forecast record. This principle proves essential for decision-making in science. The prediction model handles numerical outputs; the AQI conversion assigns categories; and the LLM’s sole role is to relay those outputs in language tailored to the user’s query. By separating these duties, each generated answer can be broken into verifiable components instead of being assessed purely on readability.

The prompt specifies a stripped-down structured input containing `predicted_pm25_ugm3`, `predicted_aqi`, `aqi_category`, and `top_features`. The `top_features` dictionary enumerates the measured feature values accessible at forecast issuance. The LLM output must reiterate the predicted value and category, pinpoint a `primary_driver` from the provided feature set, furnish a one- or two-sentence explanation, issue a `health_recommendation`, and append a `confidence_note`. Representing this as JSON facilitates independent parsing of each field: numeric entries can be cross-checked against deterministic baselines, while text fields can be examined for pollutant references or improbable numeric assertions.

The stringent audit checks adopt a deliberately conservative stance. H1 inquires not about the forecast’s accuracy but about

whether the LLM faithfully reproduced it. H2 sidesteps judging the EPA formula’s universal suitability, focusing instead on whether the LLM altered the computed AQI. H3 forestalls category mislabeling—for example, dubbing an AQI of 178 as “Moderate.” H4 blocks feature fabrication, like referencing ozone, nitrogen dioxide, sulfur dioxide, or PM10 when those variables were absent from input. H5 prohibits physically impossible numerical statements—say, a negative pressure reading or a pollutant concentration beyond the allowed range. Together, these checks transform hallucination detection into a repeatable engineering discipline.

The soft confidence metric, H6, is deliberately excluded from hard hallucination classification. Diminished token confidence may signal ambiguity or atypical wording, yet it does not automatically imply falsehood. By the same token, elevated confidence does not guarantee veracity. Hence, H6 serves primarily as a triage indicator for manual inspection. In this experiment, H6 failed to flag any explanation, with a mean average log probability of roughly -0.062. This suggests the model produced the constrained schema with considerable confidence, likely stemming from the repetitive prompt and output structure across instances.

A notable limitation of this audit lies in its emphasis on factual fidelity at the expense of practical utility. A response may satisfy H1 through H5 yet remain too vague for a particular user. Consider a recommendation to “reduce outdoor exertion”: though factually grounded, it fails to inform a school district’s decision on canceling sports. To address this, the QA analysis table incorporates qualitative interpretation. Future work should extend the audit with rubrics for actionability, readability metrics, and cross-references to a curated table of EPA-style activity advisories. A hybrid schema could employ deterministic templates for high-risk categories and reserve LLM generation for low-risk contextualization. Table VII details the audit checks.

APPENDIX C

EXTENDED FORECASTING ERROR INTERPRETATION

When interpreting forecasting results, we need to consider the time series structure. PM2.5 concentrations shift gradually over hours but can spike sharply during pollution events. Persistence works best when the next hour mirrors the current one. AR(24) brings in the previous day’s cycle, capturing recurring diurnal patterns. LSTM offers more flexibility—but that flexibility can backfire if the training objective and validation selection don’t explicitly reward rare high-pollution events. Here, both LSTM versions maintained correlation but had a noticeable negative bias at the high end.

Scatter plots reveal a classic regression-to-the-mean pattern: low PM2.5 values occasionally get overpredicted, while high ones frequently get underpredicted. That behavior helps average squared error for typical midrange cases but hurts decision-making during hazardous episodes. For air-quality managers, false reassurance during high pollution is far worse than mild over-warning during low pollution. Future studies, then, should explore category-weighted loss, threshold-specific

recall, and asymmetric penalties that put more weight on underpredicting Very Unhealthy and Hazardous hours.

The ISSA search did find a model with a larger hidden size and higher learning rate, but the validation probe only ran one epoch per candidate. That setup can favor fast learners over models that generalize well after full training—a typical pitfall in low-budget hyperparameter searches. A more robust design would evaluate candidates with early stopping over multiple epochs, align the validation metric with AQI category accuracy, and repeat the search across several random seeds. So, interpret the current result as a faithful implementation under a small compute budget, not a definitive dismissal of ISSA-LSTM. Table VIII summarizes observed issues and proposed improvements.

One final takeaway: interpretability must be linked to model quality. Even a perfectly grounded explanation of a biased forecast is still just that—biased. The LLM layer can correctly state that AQI is Unhealthy if that’s the prediction, but it can’t fix a model that consistently underestimates severe events without extra calibration or post-processing. So the safest architecture is layered: robust baseline forecasting, calibrated category mapping, uncertainty estimates, constrained explanation, and automatic auditing, each with its own metrics and failure triggers.

For deployment, an uncertainty-aware output is key. Rather than a single PM2.5 number, the system should provide a prediction interval and a category-risk distribution. Then the LLM can communicate uncertainty in plain language: “The most likely category is Unhealthy, with a non-negligible chance of Very Unhealthy conditions.” That approach aligns with how people make decisions under risk and reduces the urge to treat the LLM’s answer as a deterministic medical directive.

TABLE VII
DETAILED INTERPRETATION OF THE HALLUCINATION-AUDIT CHECKS USED FOR THE LLM EXPLANATION LAYER.

Check	Operational definition	Why it matters
H1: numeric match	Claimed PM2.5 must equal the forecast within 1%.	Prevents the LLM from rounding into a materially different forecast or substituting observed PM2.5 for predicted PM2.5.
H2: AQI integer	Claimed AQI must match deterministic AQI conversion within one point.	Prevents an explanation from using a different AQI value than the one displayed to the user.
H3: category bracket	Claimed category must match the AQI bracket.	Prevents contradictory communication such as a high AQI with a low-risk category.
H4: feature grounding	Pollutants or drivers in the explanation must belong to the supplied feature set.	Prevents fabricated causes that were not measured or passed to the model.
H5: range envelope	Numbers with units must be inside allowed physical ranges.	Detects impossible or unsupported quantitative claims in free text.
H6: confidence	Average token log probability below -2.0 is a soft warning.	Flags potentially unstable generations for review without automatically labeling them hallucinations.

TABLE VIII
ERROR-ANALYSIS SUMMARY AND PROPOSED TECHNICAL IMPROVEMENTS.

Observed issue	Evidence in this run	Recommended improvement
Underprediction of peaks	Predicted PM2.5 falls below actual values during severe episodes.	Use event-weighted loss, quantile prediction, and calibration on high-AQI ranges.
Overproduction of midrange categories	Predictions cluster in Moderate and Unhealthy bands.	Tune thresholds and calibrate category probabilities rather than relying only on point estimates.
Dominance of lagged PM2.5	RF importance gives approximately 94.5% to PM2.5 history.	Keep simple baselines and test whether meteorology improves longer-horizon forecasts.
ISSA instability	Validation loss fluctuates after hyperparameter search.	Increase search budget, use multi-epoch fitness, and report seed variability.
Generic LLM advice	Some recommendations are correct but not specific enough.	Add retrieval from official AQI guidance and policy-specific templates.
Raw JSON defects	Some raw generations are malformed before fallback parsing.	Use constrained decoding, schema validation, and repair or retry logic.

TABLE IX: Interactive QA answer analysis. Rows are derived from the generated QA examples and interpreted for grounding, actionability, and remaining limitations.

Sample	User question	Forecast context	LLM recommendation excerpt	Researcher analysis
0	Is it safe to go for a run today?	106.9 / 178 (Unhealthy)	Consider reducing or rescheduling outdoor exercise to minimize exposure to poor air quality.	Appropriately discourages prolonged outdoor exertion because the sample is Unhealthy; the answer maps AQI to activity risk rather than only restating the number.
0	Should children play outside this afternoon?	106.9 / 178 (Unhealthy)	Children should limit outdoor activities to reduce health risks.	Child-specific advice is consistent with sensitive-population framing; the answer correctly translates the same forecast into a more conservative recommendation.
0	Do I need to wear an N95 mask if I commute by bicycle?	106.9 / 178 (Unhealthy)	Consider reducing outdoor exertion and staying indoors if possible.	Commute advice is conservative; a stronger version should explicitly mention route duration, filtration, and avoiding high-exertion cycling.

Sample	User question	Forecast context	LLM recommendation excerpt	Researcher analysis
0	Is it OK to open the windows for ventilation?	106.9 / 178 (Unhealthy)	Everyone should reduce prolonged outdoor exertion; sensitive groups should avoid it.	Ventilation answer is useful but should be stricter: when outdoor PM2.5 is Unhealthy, opening windows can increase indoor particle exposure.
0	Should outdoor school sports be cancelled today?	106.9 / 178 (Unhealthy)	Sensitive groups should limit outdoor exertion.	Sports recommendation is appropriate for institutional decision support; a cancellation or rescheduling rule should be tied to local policy thresholds.
1	Is it safe to go for a run today?	67.7 / 157 (Unhealthy)	Everyone should reduce prolonged outdoor exertion; sensitive groups should avoid it.	Appropriately discourages prolonged outdoor exertion because the sample is Unhealthy; the answer maps AQI to activity risk rather than only restating the number.
1	Should children play outside this afternoon?	67.7 / 157 (Unhealthy)	Children should limit outdoor activities to reduce exposure to unhealthy air.	Child-specific advice is consistent with sensitive-population framing; the answer correctly translates the same forecast into a more conservative recommendation.
1	Do I need to wear an N95 mask if I commute by bicycle?	67.7 / 157 (Unhealthy)	Consider reducing outdoor exertion and spending time indoors when possible.	Commute advice is conservative; a stronger version should explicitly mention route duration, filtration, and avoiding high-exertion cycling.
1	Is it OK to open the windows for ventilation?	67.7 / 157 (Unhealthy)	Everyone should reduce prolonged outdoor exertion; sensitive groups should avoid it.	Ventilation answer is useful but should be stricter: when outdoor PM2.5 is Unhealthy, opening windows can increase indoor particle exposure.
1	Should outdoor school sports be cancelled today?	67.7 / 157 (Unhealthy)	Everyone should reduce prolonged outdoor exertion; sensitive groups should avoid it.	Sports recommendation is appropriate for institutional decision support; a cancellation or rescheduling rule should be tied to local policy thresholds.
2	Is it safe to go for a run today?	56.1 / 151 (Unhealthy)	Consider reducing or rescheduling outdoor exercise to protect your health.	Appropriately discourages prolonged outdoor exertion because the sample is Unhealthy; the answer maps AQI to activity risk rather than only restating the number.
2	Should children play outside this afternoon?	56.1 / 151 (Unhealthy)	Children should avoid outdoor activities to minimize health risks.	Child-specific advice is consistent with sensitive-population framing; the answer correctly translates the same forecast into a more conservative recommendation.
2	Do I need to wear an N95 mask if I commute by bicycle?	56.1 / 151 (Unhealthy)	Consider reducing outdoor exertion and staying indoors if possible.	Commute advice is conservative; a stronger version should explicitly mention route duration, filtration, and avoiding high-exertion cycling.

Sample	User question	Forecast context	LLM recommendation excerpt	Researcher analysis
2	Is it OK to open the windows for ventilation?	56.1 / 151 (Unhealthy)	Everyone should reduce prolonged outdoor exertion; sensitive groups should avoid it.	Ventilation answer is useful but should be stricter: when outdoor PM2.5 is Unhealthy, opening windows can increase indoor particle exposure.
2	Should outdoor school sports be cancelled today?	56.1 / 151 (Unhealthy)	Sensitive groups should avoid prolonged outdoor exertion.	Sports recommendation is appropriate for institutional decision support; a cancellation or rescheduling rule should be tied to local policy thresholds.
3	Is it safe to go for a run today?	30.6 / 90 (Moderate)	Unusually sensitive individuals should consider reducing prolonged exertion.	Appropriately discourages prolonged outdoor exertion because the sample is Moderate; the answer maps AQI to activity risk rather than only restating the number.
3	Should children play outside this afternoon?	30.6 / 90 (Moderate)	Unusually sensitive individuals should reduce prolonged exertion.	Child-specific advice is consistent with sensitive-population framing; the answer correctly translates the same forecast into a more conservative recommendation.
3	Do I need to wear an N95 mask if I commute by bicycle?	30.6 / 90 (Moderate)	Stay indoors if possible, especially if you have respiratory issues.	Commute advice is conservative; a stronger version should explicitly mention route duration, filtration, and avoiding high-exertion cycling.
3	Is it OK to open the windows for ventilation?	30.6 / 90 (Moderate)	Unusually sensitive individuals should reduce prolonged exertion.	Ventilation answer is useful but should be stricter: when outdoor PM2.5 is Unhealthy, opening windows can increase indoor particle exposure.
3	Should outdoor school sports be cancelled today?	30.6 / 90 (Moderate)	Unusually sensitive individuals should consider reducing prolonged exertion.	Sports recommendation is appropriate for institutional decision support; a cancellation or rescheduling rule should be tied to local policy thresholds.
4	Is it safe to go for a run today?	56.0 / 151 (Unhealthy)	Everyone should reduce prolonged outdoor exertion; sensitive groups should avoid it.	Appropriately discourages prolonged outdoor exertion because the sample is Unhealthy; the answer maps AQI to activity risk rather than only restating the number.
4	Should children play outside this afternoon?	56.0 / 151 (Unhealthy)	Children should avoid outdoor activities to minimize respiratory risks.	Child-specific advice is consistent with sensitive-population framing; the answer correctly translates the same forecast into a more conservative recommendation.
4	Do I need to wear an N95 mask if I commute by bicycle?	56.0 / 151 (Unhealthy)	Consider reducing outdoor physical activity and use an N95 mask for necessary outings.	Commute advice is conservative; a stronger version should explicitly mention route duration, filtration, and avoiding high-exertion cycling.

Sample	User question	Forecast context	LLM recommendation excerpt	Researcher analysis
4	Is it OK to open the windows for ventilation?	56.0 / 151 (Unhealthy)	Everyone should reduce prolonged outdoor exertion; sensitive groups should avoid it.	Ventilation answer is useful but should be stricter: when outdoor PM2.5 is Unhealthy, opening windows can increase indoor particle exposure.
4	Should outdoor school sports be cancelled today?	56.0 / 151 (Unhealthy)	Everyone should reduce prolonged outdoor exertion; sensitive groups should avoid it.	Sports recommendation is appropriate for institutional decision support; a cancellation or rescheduling rule should be tied to local policy thresholds.